

Comparison of vitamin D2 and vitamin D3 supplementation in raising serum 25-hydroxyvitamin D status: a systematic review and meta-analysis

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Background: Currently, there is a lack of clarity in the literature as to whether there is a definitive difference between the effects of vitamins D2 and D3 in the raising of serum 25-hydroxyvitamin D [25(OH)D]. Objective: The objective of this article was to report a systematic review and meta-analysis of randomized controlled trials (RCTs) that have directly compared the effects of vitamin D2 and vitamin D3 on serum 25(OH)D concentrations in humans. Design: The ISI Web of Knowledge (January 1966 to July 2011) database was searched electronically for all relevant studies in adults that directly compared vitamin D3 with vitamin D2. The Cochrane Clinical Trials Registry, International Standard Randomized Controlled Trials Number register, and clinicaltrials.gov were also searched for any unpublished trials. Results: A meta-analysis of RCTs indicated that supplementation with vitamin D3 had a significant and positive effect in the raising of serum 25(OH)D concentrations compared with the effect of vitamin D2 ($P = 0.001$). When the frequency of dosage administration was compared, there was a significant response for vitamin D3 when given as a bolus dose ($P = 0.0002$) compared with administration of vitamin D2, but the effect was lost with daily supplementation. Conclusions: This meta-analysis indicates that vitamin D3 is more efficacious at raising serum 25(OH)D concentrations than is vitamin D2, and thus vitamin D3 could potentially become the preferred choice for supplementation. However, additional research is required to examine the metabolic pathways involved in oral and intramuscular administration of vitamin D and the effects across age, sex, and ethnicity, which this review was unable to verify.